

Control Systems I Exercise

About me: ▫ 1st year in MSc Robotics

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▫ TA also in Dynamics (LAB 659)



CS 1 Exercise ML J37.1

WhatsApp group



Course material: ▫ Moodle : Lecture Slides & Exercise Sheets
▫ Jupyter Notebooks / Simulations
▫ Lecture Notes
▫ CS I GPT

Recital structure: 1. Theory recap
2. Examples / Applications
3. Time for questions / indep. PS solving

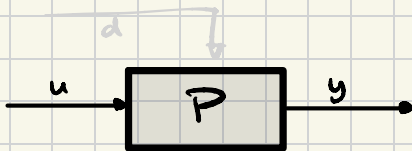
Based on:

- LinAlg II → ODEs
- Dynamics → Modeling
- Analysis III → Laplace Transforms
- (▫ Elektrotechnik I → Transfer funct.)

⇒ very abstract course BUT it will eventually make sense !

Control Systems Basics

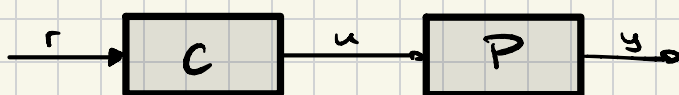
Open loop:



u = input
 y = output
 P = plant / system
 d = disturbances

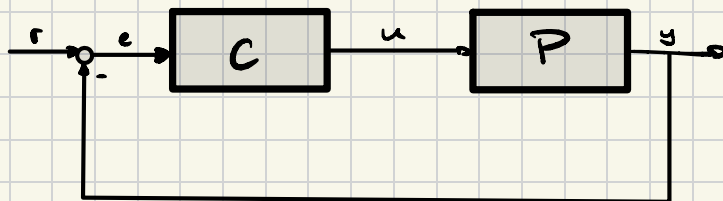
} single input single output (SISO)

e.g. driving a car
input - driver
plant - car
output - or



r = reference
 C = controller

Closed loop:



$e = r - y$ = error

Difference

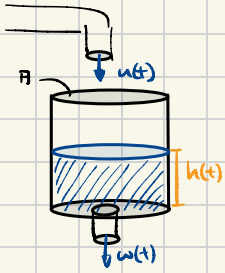
OL : i know car goes 50 km/h while pressing 30% of pedal
CL : look at error ⇒ find needed correction

- ⇒ Control objectives:
1. **Stabilization** : little deviation from reference ($e \rightarrow 0$)
 2. **Performance** : execute as accurately as possible (quick)
 3. **Robustness** : good performance even if disturbances are present

Course objectives

1. **Modeling**: use Dynamics $\Rightarrow$ represent them in Model $\rightarrow$ state space model
2. **Analysis**: understand ^{input} ^{output} I/O relation $\rightarrow$ predict behaviour
3. **Synthesis**: change system from feedback

Modeling example



$u(t)$ = incoming flow $[\frac{m^3}{s}]$
 $w(t)$ = outgoing flow $[\frac{m^3}{s}]$
 $h(t)$ = water level $[m]$
 A = cross-sectional area $[m^2]$

$$\left. \begin{array}{l} \text{Model} \\ A \cdot \frac{dh}{dt} = u - w \\ y = h \end{array} \right\}$$

$\Rightarrow$ represent with matrices $\Rightarrow$ state-space model
(just like LinAlg)